1. Majority Voting

Formula:

The predicted class $\hat{y}$ is the class with the highest frequency among the K nearest neighbors:

$$\hat{y} = \text{argmax}_{c \in C} \sum_{i=1}^{K} \mathbb{I}(y_i = c)$$

where C is the set of classes, y_i is the class label of the i-th nearest neighbor, and $\mathbb{I}$ is the indicator function (1 if true, 0 otherwise).

Numeric Example:

Suppose $K = 5$ nearest neighbors have labels: $ClassA, ClassA, ClassB, ClassA, ClassB$.

Count for Class A: 3

Count for Class B: 2

The predicted class is **Class A** (highest count).

2. Distance Weighted Voting

Formula:

Weights are assigned inversely proportional to distance (e.g., $1/d$). The predicted class is the class with the highest weighted sum:

$$\hat{y} = \text{argmax}_{c \in C} \sum_{i=1}^{K} \frac{\mathbb{I}(y_i = c)}{d_i}$$

where d_i is the distance from the target point to the i-th nearest neighbor.

Numeric Example:

Suppose $K = 3$ nearest neighbors with distances and labels:

Neighbor 1: $d_1 = 1.0$, Class A

Neighbor 2: $d_2 = 2.0$, Class B

Neighbor 3: $d_3 = 0.5$, Class A

Calculate weighted sums:

Class A: $\frac{1}{1.0} + \frac{1}{0.5} = 1 + 2 = 3$

Class B: $\frac{1}{2.0} = 0.5$

The predicted class is **Class A** (highest weighted sum).

3. Kernel Smoothing

Formula:

A Gaussian kernel (or other function) weights distances nonlinearly. The predicted class is the class with the highest kernel-weighted sum:

$$\hat{y} = \text{argmax}_{c \in C} \sum_{i=1}^{K} \mathbb{I}(y_i = c) \cdot \exp\left(-\frac{d_i^2}{2\sigma^2}\right)$$

where σ is the kernel bandwidth (controls smoothness).

Numeric Example:

Suppose $K = 3$ nearest neighbors with distances and labels, and $\sigma = 1.0$:

Neighbor 1: $d_1 = 1.0$, Class A

Neighbor 2: $d_2 = 2.0$, Class B

Neighbor 3: d_3 = *0.5*, Class A

Calculate kernel weights (Gaussian):

Neighbor 1: $\exp\left(-\frac{1.0^2}{2\cdot 1.0^2}\right) = \exp(-0.5) \approx 0.6065$

Neighbor 2: $\exp\left(-\frac{2.0^2}{2\cdot 1.0^2}\right) = \exp(-2) \approx 0.1353$

Neighbor 3: $\exp\left(-\frac{0.5^2}{2\cdot 1.0^2}\right) = \exp(-0.125) \approx 0.8825$

Calculate kernel-weighted sums:

Class A: $0.6065 + 0.8825 = 1.489$

Class B: 0.1353

The predicted class is **Class A** (highest kernel-weighted sum).